Supplemental Material: Electromagnetic Unification in the Σ-Framework

Sheng-Kai Huang[1,*]

[1] *Independent Researcher*

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We provide detailed proofs for four results connecting electromagnetism to the Σ-framework developed in Papers I–IV [1–4]. Section S1 proves that the Σ-framework uniquely selects complex Hilbert space and hence $U(1)$ gauge symmetry, excluding real and quaternionic alternatives. Section S2 establishes the DBI–Σ correspondence, showing that the Born–Infeld Lagrangian and the Khronon ghost-condensation kinetic function are two instances of a single $\Sigma = -\ln\det(\mathbb{I} - M)$ structure. Section S3 derives modified Maxwell equations from Kaluza–Klein reduction with a Σ-dependent dilaton, obtaining the gravitationally coupled field equation $\nabla_\nu F^{\mu\nu} + 3\beta(\nabla_\nu \Sigma_{\text{grav}})F^{\mu\nu} = 0$ and its observational constraints. Section S4 demonstrates the decomposition $\Sigma_{\text{total}} = \Sigma_{\text{grav}} + \Sigma_{\text{EM}}$ from the five-dimensional Kaluza–Klein effective lapse, with explicit cross-coupling and the formal analogy between α-running and Σ_{grav}. All proofs use standard results from quantum information theory, algebraic QFT, and Kaluza–Klein geometry; no new axioms are introduced.

CONTENTS

I. S1. $U(1)$ UNIQUENESS FROM THE Σ-FRAMEWORK

In this section we prove that the quantum relative entropy (QRE) $D(\rho\|\sigma)$, and hence the entropy production $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$, *requires* complex Hilbert space and uniquely selects $U(1)$ as the minimal continuous gauge symmetry. The argument proceeds in three steps: complex is necessary (Theorem I.1), quaternionic is excluded (Theorem I.3), and $U(1)$ follows uniquely (Corollary I.4).

* akai@fawstudio.com

A. Complex Hilbert space from QRE

Theorem I.1 (Complex Hilbert Space from QRE)**.** *The Araki relative entropy $D_{\text{Araki}}(\omega\|\omega_0)$ [6] is well-defined only on complex Hilbert spaces. Specifically, its definition requires the modular operator $\Delta_{\omega_0,\omega}$ from the Tomita–Takesaki modular theory, whose one-parameter automorphism group $\sigma_t(A) = \Delta^{it} A \Delta^{-it}$ necessitates complex exponentiation.*

Proof. The Araki relative entropy is defined as [6]

$$D_{\text{Araki}}(\omega\|\omega_0) = -\langle \Omega,\ \big(\ln \Delta_{\omega_0,\omega}\big)\Omega \rangle, \tag{1}$$

where $\Delta_{\omega_0,\omega}$ is the relative modular operator and Ω is the vector representative of ω in the natural cone.

The Tomita–Takesaki theorem [7, 8] constructs the modular operator as follows. Given a von Neumann algebra $\mathcal{M}$ acting on a Hilbert space $\mathcal{H}$ with a cyclic and separating vector Ω, the Tomita operator S_0 is defined by

$$S_0\, A\Omega = A^*\Omega, \qquad A \in \mathcal{M}. \tag{2}$$

Its polar decomposition $S_0 = J\Delta^{1/2}$ yields the modular operator $\Delta = S_0^* S_0$ (positive, generally unbounded) and the modular conjugation J (antilinear isometry). The modular automorphism group is

$$\sigma_t(A) = \Delta^{it} A\, \Delta^{-it}, \qquad t \in \mathbb{R}. \tag{3}$$

The operator Δ^{it} is defined via the spectral theorem: if $\Delta = \int_0^\infty \lambda\, dE(\lambda)$, then

$$\Delta^{it} = \int_0^\infty \lambda^{it}\, dE(\lambda) = \int_0^\infty e^{it\ln\lambda}\, dE(\lambda). \tag{4}$$

The exponent $e^{it\ln\lambda}$ is a complex phase $e^{i\theta}$ with $\theta = t\ln\lambda \in \mathbb{R}$.

On a *real* Hilbert space $\mathcal{H}_{\mathbb{R}}$, the spectral theorem still holds for self-adjoint operators, but the phase $e^{i\theta}$ is not a real number for generic θ. The operator Δ^{it} would map vectors out of $\mathcal{H}_{\mathbb{R}}$, since $e^{i\theta}\psi \notin \mathcal{H}_{\mathbb{R}}$ unless $\theta \in \pi\mathbb{Z}$. Therefore:

1. The modular automorphism group σ_t cannot be defined as a continuous one-parameter group on $\mathcal{H}_{\mathbb{R}}$.
2. Without σ_t, the relative modular operator $\Delta_{\omega_0,\omega}$ and its logarithm are not available.
3. Without $\ln \Delta_{\omega_0,\omega}$, the Araki relative entropy (1) cannot be defined.

Since the Σ-framework is built on $\Sigma = D_{\mathrm{Araki}}(\omega|_{\mathcal{A}} \| \omega_{\mathrm{ref}}|_{\mathcal{A}})$ [4], the entire framework requires $\mathcal{H}$ to be complex. □

Remark. *For finite-dimensional systems the Umegaki QRE $D(\rho\|\sigma) = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma)]$ [5] also requires complex structure: the matrix logarithm of a density operator involves the eigendecomposition $\rho = \sum_i p_i |i\rangle\langle i|$, which on a real Hilbert space would restrict to symmetric matrices, excluding the full space of quantum states (e.g., any state involving coherences with complex phases).*

B. Poincaré symmetry forces complex structure

Theorem I.2 (Moretti–Oppio [12]). *Let $\mathcal{H}$ be a separable Hilbert space over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and let $U : \widetilde{\mathcal{P}}_+^{\uparrow} \to \mathcal{U}(\mathcal{H})$ be an irreducible, strongly continuous, unitary representation of the universal cover of the proper orthochronous Poincaré group satisfying the spectral condition (non-negative energy). Then $\mathbb{K} = \mathbb{C}$, or $\mathcal{H}$ admits a unique $\mathbb{C}$-structure $J_{\mathbb{C}}$ commuting with U such that $(\mathcal{H}, J_{\mathbb{C}})$ is a complex Hilbert space on which U acts as a standard complex-unitary representation.*

In other words, the requirement of Poincaré covariance with positive energy uniquely selects complex quantum mechanics. Real Hilbert spaces fail because they cannot accommodate the necessary phase structure; quaternionic Hilbert spaces are automatically reducible to complex ones under Poincaré [12].

This result is independent of Theorem I.1 and provides a second, symmetry-based argument for $\mathbb{K} = \mathbb{C}$.

C. Tensor product excludes quaternionic

Theorem I.3 (Tensor Product Exclusion). *The Petz recovery map $\mathcal{R}_{\sigma,\mathcal{N}}$ [9, 10] requires the tensor product decomposition $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ for the system–environment split. Quaternionic Hilbert spaces do not admit a natural tensor product, and therefore the Petz map—and hence the Σ-framework—cannot be formulated over $\mathbb{H}$.*

Proof. The Petz recovery map for a channel $\mathcal{N}$ with Stinespring dilation $\mathcal{N}(\rho) = \mathrm{Tr}_E[V\rho V^\dagger]$, where $V : \mathcal{H}_S \to \mathcal{H}_S \otimes \mathcal{H}_E$, is defined as [10, 11]

$$\mathcal{R}_{\sigma,\mathcal{N}}(\cdot) = \sigma^{1/2}\, \mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\,(\cdot)\,\mathcal{N}(\sigma)^{-1/2}\big)\,\sigma^{1/2}, \quad (5)$$

where $\mathcal{N}^\dagger$ is the adjoint channel. This construction requires:

1. A tensor product $\mathcal{H}_S \otimes \mathcal{H}_E$ for the Stinespring dilation,
2. A partial trace Tr_E over $\mathcal{H}_E$,
3. The adjoint map $\mathcal{N}^\dagger$ defined via $\mathrm{Tr}[A\,\mathcal{N}(B)] = \mathrm{Tr}[\mathcal{N}^\dagger(A)\,B]$.

Adler [13] showed that quaternionic Hilbert spaces lack a canonical tensor product: if $\mathcal{H}_1$ and $\mathcal{H}_2$ are right $\mathbb{H}$-modules, the naive construction $\mathcal{H}_1 \otimes_{\mathbb{H}} \mathcal{H}_2$ collapses to a *real* vector space (not a quaternionic module), because the quaternionic scalars do not commute with the tensor product structure. Specifically, for $q \in \mathbb{H}$,

$$(\psi q) \otimes \phi \neq \psi \otimes (q\phi) \quad (6)$$

in general, since q does not commute with the elements of $\mathcal{H}_2$. This non-commutativity prevents the definition of a well-behaved partial trace.

Without tensor products:

- No Stinespring dilation $\Rightarrow$ no purification of channels,
- No partial trace $\Rightarrow$ no reduced states,
- No Petz map $\Rightarrow$ no recovery fidelity F,
- No $F \Rightarrow$ no $\tau = 1 - F \Rightarrow$ no Σ.

The quaternionic option is therefore structurally incompatible with the Σ-framework. □

D. $U(1)$ as the unique gauge symmetry

Corollary I.4 ($U(1)$ Uniqueness). *The Σ-framework uniquely selects $U(1)$ as the minimal continuous gauge symmetry of the underlying Hilbert space.*

Proof. By Theorems I.1 and I.3, the Σ-framework requires $\mathcal{H}$ over $\mathbb{C}$. On a complex Hilbert space, physical states are rays $[\psi] = \{e^{i\theta}\psi : \theta \in \mathbb{R}\}$. The group of transformations $\psi \mapsto e^{i\theta}\psi$ that preserves the ray space is $U(1)$, acting as a global phase rotation.

Promoting this global $U(1)$ to a local symmetry $\psi(x) \mapsto e^{i\theta(x)}\psi(x)$ via the standard gauge principle [14] requires a connection A_μ transforming as

$$A_\mu \mapsto A_\mu + \partial_\mu\theta, \quad (7)$$

with field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. This is precisely the structure of electromagnetism.

The uniqueness follows from the chain:

- Σ requires complex $\mathcal{H}$ (Theorem I.1),
- Complex $\mathcal{H}$ excludes real and quaternionic (Theorem I.2, I.3),

- Complex scalars have automorphism group $\mathrm{Aut}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}_2$ (complex conjugation), but the continuous symmetry of the ray space is $U(1)$,
- $U(1)$ is the *unique* connected compact Lie group of dimension one.

Therefore, the Σ-framework, through its requirement of complex Hilbert space, uniquely selects $U(1)$ gauge symmetry—the gauge group of electromagnetism. □

Remark. *This does not derive $SU(3)\times SU(2)\times U(1)$; it derives only the $U(1)_{\mathrm{EM}}$ factor. The non-abelian gauge groups require additional structure (e.g., internal degrees of freedom from compactification or flavor symmetry) beyond the minimal requirements of the Σ-framework.*

II. S2. DBI–Σ CORRESPONDENCE

We now show that the Born–Infeld (BI) Lagrangian of nonlinear electrodynamics and the DBI kinetic function of the Khronon field are two instances of a single $\Sigma = -\ln\det(\mathbb{I} - M)$ structure.

A. Σ for DBI kinetic function

Lemma II.1 (Σ for DBI Kinetic Function)**.** *Consider the DBI extension of the ghost-condensation kinetic function [17, 18]:*

$$K_{\mathrm{DBI}}(Q) = \frac{2\mu^2}{\lambda_D}\Big[1 - \sqrt{1 - \lambda_D(Q-1)^2}\,\Big], \tag{8}$$

where $Q = 1/(N_\phi)$ is the inverse Khronon lapse, μ is the Khronon mass parameter, and λ_D controls the DBI scale. Define $\delta \equiv Q - 1$. The effective Khronon lapse is

$$N_\phi^{\mathrm{eff}} = \sqrt{1 - \lambda_D\delta^2}. \tag{9}$$

Then the entropy production is

$$\Sigma_{\mathrm{DBI}} = -2\ln N_\phi^{\mathrm{eff}} = -\ln(1 - \lambda_D\delta^2). \tag{10}$$

Proof. From Paper II [2], the gravitational entropy production for a background with effective lapse N_ϕ is $\Sigma = -2\ln N_\phi = 2\ln Q$.

For the DBI kinetic function (8), the effective lapse receives a correction from the nonlinear kinetic structure. The DBI action can be written as

$$S_{\mathrm{DBI}} = \int \sqrt{-g}\,\frac{2\mu^2}{\lambda_D}\big[1 - \sqrt{1 - \lambda_D\delta^2}\,\big]\,d^4x. \tag{11}$$

The canonical momentum conjugate to δ is

$$\pi_\delta = \frac{\partial K_{\mathrm{DBI}}}{\partial\dot{\delta}} = \frac{2\mu^2\,\delta}{\sqrt{1 - \lambda_D\delta^2}}, \tag{12}$$

and the effective sound speed is bounded by $c_s^2 = (1 - \lambda_D\delta^2)$.

The effective lapse governing the causal structure of Khronon perturbations is therefore $N_\phi^{\mathrm{eff}} = \sqrt{1 - \lambda_D\delta^2}$, and by the standard identification $\Sigma = -2\ln N_\phi$:

$$\Sigma_{\mathrm{DBI}} = -2\ln\sqrt{1 - \lambda_D\delta^2} = -\ln(1 - \lambda_D\delta^2). \tag{13}$$

For $\lambda_D\delta^2 \ll 1$, this reduces to $\Sigma_{\mathrm{DBI}} \approx \lambda_D\delta^2 \approx 2\ln(1+\delta)$ at leading order, recovering the ghost-condensation result $\Sigma = 2\ln Q$. □

B. Born–Infeld as Σ_{EM}

Theorem II.2 (Born–Infeld as Electromagnetic Σ)**.** *The Born–Infeld (BI) Lagrangian [15]*

$$\mathcal{L}_{\mathrm{BI}} = b^2\left(1 - \sqrt{1 + \frac{F_{\mu\nu}F^{\mu\nu}}{2b^2}}\right) \tag{14}$$

(Lorentzian signature $(-+++)$, with b the critical field strength) defines an electromagnetic entropy production

$$\Sigma_{\mathrm{BI}} = -\ln\left(1 - \frac{E^2}{b^2}\right) \tag{15}$$

in the pure electric limit ($\boldsymbol{B} = 0$, $E < b$).

Proof. The BI effective metric governing photon propagation in the BI medium is [15, 16]

$$\tilde{g}^{\mu\nu} = \eta^{\mu\nu} + \frac{F^{\mu\alpha}F_\alpha{}^\nu}{b^2} + \ldots \tag{16}$$

For a uniform electric field $\boldsymbol{E} = E\hat{z}$ with $\boldsymbol{B} = 0$, the field-strength invariant is $F_{\mu\nu}F^{\mu\nu} = -2E^2$ (in our conventions), so

$$1 + \frac{F_{\mu\nu}F^{\mu\nu}}{2b^2} = 1 - \frac{E^2}{b^2}. \tag{17}$$

The BI Lagrangian becomes $\mathcal{L}_{\mathrm{BI}} = b^2(1 - \sqrt{1 - E^2/b^2})$.

The determinant of the BI effective metric in the pure electric case reduces to [16]

$$\det\left(\eta_{\mu\nu} + \frac{F_{\mu\nu}}{b}\right) = 1 - \frac{E^2}{b^2} \tag{18}$$

(in the $\tilde{F} = 0$ sector, where $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}$ is the dual).

We define the electromagnetic entropy production by analogy with the gravitational case $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$:

$$\Sigma_{\mathrm{BI}} \equiv -\ln\det(\mathbb{I} - M_{\mathrm{EM}}), \tag{19}$$

where $M_{\mathrm{EM}} = E^2/b^2$ in the pure electric limit. Substituting (18):

$$\Sigma_{\mathrm{BI}} = -\ln\left(1 - \frac{E^2}{b^2}\right). \tag{20}$$

This is well-defined for $E < b$ (sub-critical field) and diverges as $E \to b$, signaling maximal information loss at the BI critical field—the electromagnetic analogue of the event horizon where $\Sigma_{\mathrm{grav}} \to \infty$. □

C. Unified $\Sigma = -\ln\det(\mathbb{I} - M)$

Theorem II.3 (Unified Determinant Formula)**.** *Both the gravitational and electromagnetic entropy productions are special cases of*

$$\boxed{\Sigma = -\ln\det(\mathbb{I} - M)} \tag{21}$$

with the following identifications:

1. ***Gravity (DBI Khronon):*** $M_{\text{grav}} = \lambda_D \delta^2$*, where* $\delta = Q - 1$ *is the Khronon displacement and* λ_D *is the DBI scale parameter. Then* $\Sigma_{\text{DBI}} = -\ln(1 - \lambda_D \delta^2)$.
2. ***Electromagnetism (Born–Infeld):*** $M_{\text{EM}} = E^2/b^2$*, where* E *is the electric field strength and* b *is the BI critical field. Then* $\Sigma_{\text{BI}} = -\ln(1 - E^2/b^2)$.
3. ***Combined (leading order):*** *In a system with both gravitational and electromagnetic fields,*

$$\Sigma_{\text{total}} = -\ln\det(\mathbb{I} - M_{\text{grav}} - M_{\text{EM}}) \tag{22}$$

to leading order in the cross-coupling.

Proof. Cases (1) and (2) follow directly from Lemma II.1 and Theorem II.2.

For case (3), the structural parallel is exact: in both cases, Σ measures the departure of an effective metric determinant from unity. When both deformations are present, the combined effective metric is schematically

$$g^{\text{eff}}_{\mu\nu} = \eta_{\mu\nu} + h^{\text{grav}}_{\mu\nu} + h^{\text{EM}}_{\mu\nu}, \tag{23}$$

and to leading order in both perturbations (i.e., $\lambda_D \delta^2 \ll 1$ and $E^2/b^2 \ll 1$),

$$\begin{aligned} \Sigma_{\text{total}} &= -\ln\det\big(\mathbb{I} - M_{\text{grav}} - M_{\text{EM}}\big) \\ &\approx M_{\text{grav}} + M_{\text{EM}} \\ &= \Sigma_{\text{grav}} + \Sigma_{\text{EM}} \end{aligned} \tag{24}$$

at linear order, with cross-terms appearing at $O(M_{\text{grav}} \cdot M_{\text{EM}})$. The exact non-perturbative formula (22) includes all cross-couplings. □

Remark. *The identification* $\lambda_D \delta^2 \leftrightarrow E^2/b^2$ *makes precise the analogy between the Khronon field and the electromagnetic field within the* Σ*-framework:*

	Gravity	***EM***
Field variable	$\delta = Q - 1$	E/b
Critical scale	$\lambda_D^{-1/2}$	b
Σ *divergence*	$\lambda_D \delta^2 \to 1$	$E \to b$
Physical limit	*horizon*	*critical field*

Both represent information-theoretic horizons beyond which the Petz recovery fidelity $F = e^{-\Sigma/2} \to 0$.

III. S3. MODIFIED MAXWELL EQUATIONS

We derive the gravitationally modified Maxwell equations from Kaluza–Klein (KK) reduction with a Σ-dependent dilaton, and obtain observational constraints.

A. KK field equations with dynamic dilaton

Theorem III.1 (KK Field Equations with Dynamic Dilaton)**.** *Consider the five-dimensional Kaluza–Klein metric*

$$ds_5^2 = g_{\mu\nu}\, dx^\mu dx^\nu + f^2 \big(dy + A_\mu\, dx^\mu\big)^2, \tag{25}$$

where $g_{\mu\nu}$ *is the four-dimensional spacetime metric,* A_μ *is the gauge potential,* f *is the dilaton (radius of the compact dimension), and* $y \sim y + 2\pi R$ *is the compact coordinate. The five-dimensional vacuum Einstein equations* $R^{(5)}_{MN} = 0$ *yield, from the* $(\mu, 5)$ *component, the modified Maxwell equation*

$$\nabla_\nu \big(f^3 F^{\mu\nu}\big) = 0, \tag{26}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$*. Expanding via the Leibniz rule:*

$$\boxed{\nabla_\nu F^{\mu\nu} + 3\,(\nabla_\nu \ln f)\, F^{\mu\nu} = 0.} \tag{27}$$

Proof. The five-dimensional metric (25) in matrix form is

$$G^{(5)}_{MN} = \begin{pmatrix} g_{\mu\nu} + f^2 A_\mu A_\nu & f^2 A_\mu \\ f^2 A_\nu & f^2 \end{pmatrix}. \tag{28}$$

The five-dimensional Ricci tensor components are (see, e.g., Refs. [19, 20]):

$$R^{(5)}_{\mu\nu} = R^{(4)}_{\mu\nu} - \frac{f^2}{2}\, F_{\mu\alpha} F_\nu{}^\alpha - \frac{1}{f} \nabla_\mu \nabla_\nu f, \tag{29}$$

$$R^{(5)}_{\mu 5} = \frac{f}{2}\, \nabla_\nu \big(f^2 F^{\mu\nu}\big) = \frac{1}{2}\, \nabla_\nu \big(f^3 F^{\mu\nu}\big) \cdot \frac{1}{f}, \tag{30}$$

where in the second line we used $\nabla_\nu(f^2 F^{\mu\nu}) = f^{-1} \nabla_\nu (f^3 F^{\mu\nu}) - f^2 F^{\mu\nu} \nabla_\nu \ln f$. More directly, the standard KK reduction gives the $(\mu, 5)$ equation as [19]

$$R^{(5)}_{\mu 5} = 0 \quad \Longrightarrow \quad \nabla_\nu \big(f^3 F^{\mu\nu}\big) = 0. \tag{31}$$

Expanding with the Leibniz rule:

$$\begin{aligned} 0 &= \nabla_\nu \big(f^3 F^{\mu\nu}\big) \\ &= f^3\, \nabla_\nu F^{\mu\nu} + 3 f^2\, (\nabla_\nu f)\, F^{\mu\nu} \\ &= f^3 [\nabla_\nu F^{\mu\nu} + 3\,(\nabla_\nu \ln f)\, F^{\mu\nu}]\,. \end{aligned} \tag{32}$$

Since $f > 0$, we may divide by f^3 to obtain Eq. (27). □

Remark. *The coefficient 3 in front of $\nabla_\nu \ln f$ arises from the f^3 factor in the KK reduction, which in turn comes from the $\sqrt{-G^{(5)}} = f\sqrt{-g^{(4)}}$ volume element combined with the gauge kinetic term structure. A common approximation uses f^2 (coefficient 2), but the rigorous five-dimensional reduction yields f^3.*

Now we connect the dilaton to the gravitational Σ.

Proposition III.2 (Σ-Dependent Dilaton)**.** *If the dilaton is a function of the gravitational entropy production,*

$$f = f_0 \exp(\beta\, \Sigma_{\rm grav}), \tag{33}$$

where β is a dimensionless coupling constant and $\Sigma_{\rm grav} = -\ln(-g_{00})$ [2], then the modified Maxwell equation (27) becomes

$$\nabla_\nu F^{\mu\nu} + 3\beta\,(\nabla_\nu \Sigma_{\rm grav})\, F^{\mu\nu} = 0. \tag{34}$$

Proof. From Eq. (33), $\nabla_\nu \ln f = \beta\, \nabla_\nu \Sigma_{\rm grav}$. Substituting into Eq. (27) yields Eq. (34) immediately. $\square$

B. Modified Coulomb law

Corollary III.3 (Modified Coulomb Law)**.** *On a Schwarzschild background with $\Sigma_{\rm grav} = -\ln(1 - r_{\rm s}/r)$, the modified Maxwell equation (34) yields, for a static point charge Q_e, the modified Coulomb potential*

$$\Phi(r) = \frac{Q_e}{4\pi r}\left[1 + \beta'\,\frac{r_{\rm s}}{r}\,\ln\!\left(\frac{r}{r_{\rm s}}\right)\right], \tag{35}$$

where $\beta' = 3\beta$ and $r_{\rm s} = 2G_{\rm N}M/c^2$ is the Schwarzschild radius.

Proof. On the Schwarzschild background, the metric component is $-g_{00} = 1 - r_{\rm s}/r$, so $\Sigma_{\rm grav} = -\ln(1 - r_{\rm s}/r)$. The radial gradient is

$$\nabla_r \Sigma_{\rm grav} = \frac{\partial}{\partial r}\big[-\ln(1 - r_{\rm s}/r)\big] = \frac{r_{\rm s}}{r(r - r_{\rm s})}. \tag{36}$$

For $r \gg r_{\rm s}$, this simplifies to $\nabla_r \Sigma_{\rm grav} \approx r_{\rm s}/r^2$.

The static, spherically symmetric Maxwell equation with the Σ-coupling (34) reduces (in the radial direction, for the electric field $E_r = -\partial_r \Phi$) to

$$\frac{1}{r^2}\frac{d}{dr}\big(r^2 E_r\big) + 3\beta\,\frac{r_{\rm s}}{r(r - r_{\rm s})}\,E_r = \frac{Q_e}{\epsilon_0}\,\delta^3(\boldsymbol{r}). \tag{37}$$

In the weak-field regime $r \gg r_{\rm s}$, we solve perturbatively. Write $E_r = E_r^{(0)} + \beta'\, E_r^{(1)}$ with $\beta' = 3\beta$. The zeroth order gives $E_r^{(0)} = Q_e/(4\pi r^2)$. The first-order correction satisfies

$$\frac{1}{r^2}\frac{d}{dr}\Big(r^2 E_r^{(1)}\Big) = -\frac{r_{\rm s}}{r^2}\,\frac{Q_e}{4\pi r^2}, \tag{38}$$

which integrates to

$$E_r^{(1)} = \frac{Q_e}{4\pi r^2}\,\frac{r_{\rm s}}{r}\,\ln\!\left(\frac{r}{r_{\rm s}}\right). \tag{39}$$

The corresponding potential (integrating $-E_r$ with boundary condition $\Phi \to 0$ as $r \to \infty$) gives Eq. (35) to leading order in $r_{\rm s}/r$. $\square$

C. Binary pulsar constraint

Example III.1 (Binary Pulsar Constraint)**.** *The double pulsar PSR J0737−3039 [21] provides a strong-field test of the modified Coulomb law (35).*

The orbital parameters of this system are:

- Orbital separation: $a \approx 8.8 \times 10^8$ m,
- Pulsar mass: $M_A \approx 1.34\, M_\odot$,
- Schwarzschild radius: $r_{\rm s} \approx 3.96$ km,
- Compactness: $r_{\rm s}/a \approx 4.5 \times 10^{-6}$, but at closest approach $r_{\rm min} \approx 0.7a$, the effective $r_{\rm s}/r \sim 0.3$ for surface effects.

The timing precision of PSR J0737−3039 constrains deviations from general relativity at the $\sim 10^{-3}$ level [21, 22]. The correction in (35) enters the orbital dynamics through the electromagnetic energy of charged plasma in the magnetosphere. Demanding that the Σ-correction does not exceed the timing residuals:

$$|\beta'|\,\frac{r_{\rm s}}{r}\,\left|\ln\!\left(\frac{r}{r_{\rm s}}\right)\right| < 10^{-3}, \tag{40}$$

with $r_{\rm s}/r \sim 0.3$ at the neutron star surface and $\ln(r/r_{\rm s}) \sim 1$, this gives

$$\boxed{|\beta| < 10^{-3}.} \tag{41}$$

This is the leading observational constraint on the dilaton–Σ coupling.

Remark. *The bound (41) applies to the* effective *coupling β of the KK dilaton. If $\beta = 0$ identically (i.e., the dilaton is constant and the extra dimension does not couple to $\Sigma_{\rm grav}$), Maxwell's equations are unmodified and the Σ-framework reproduces standard electrodynamics exactly. This is the minimal and most conservative option.*

IV. S4. $\Sigma_{\rm total}$ DECOMPOSITION

A. Five-dimensional lapse decomposition

Theorem IV.1 ($\Sigma_{\rm total}$ Decomposition from KK Lapse)**.** *The total entropy production in the Kaluza–Klein framework decomposes as*

$$\Sigma_{\rm total} = \Sigma_{\rm grav} + \Sigma_{\rm EM}, \tag{42}$$

where

$$\Sigma_{\rm grav} = -\ln(-g_{00}), \tag{43}$$

$$\Sigma_{\rm EM} = -\ln\!\left(1 - \frac{f^2 A_0^2}{|g_{00}|}\right), \tag{44}$$

and $\Sigma_{\rm EM}$ depends explicitly on $\Sigma_{\rm grav}$ through the factor $|g_{00}|^{-1} = e^{\Sigma_{\rm grav}}$.

Proof. The five-dimensional KK metric (25) has the $(0,0)$ component:

$$G_{00}^{(5)} = g_{00} + f^2 A_0^2. \tag{45}$$

The five-dimensional lapse is defined by $-G_{00}^{(5)} = -g_{00} - f^2 A_0^2$, and the total entropy production is

$$\begin{aligned}\Sigma_{\text{total}} &= -\ln\bigl(-G_{00}^{(5)}\bigr)\\ &= -\ln\bigl(-g_{00} - f^2 A_0^2\bigr)\\ &= -\ln(-g_{00}) - \ln\left(1 + \frac{f^2 A_0^2}{-g_{00}}\right)\\ &= \Sigma_{\text{grav}} - \ln\left(1 - \frac{f^2 A_0^2}{|g_{00}|}\right),\end{aligned} \tag{46}$$

where in the last line we used $-g_{00} = |g_{00}|$ (for signature $(-{+}{+}{+})$), and the sign in the logarithm becomes negative because $f^2 A_0^2 > 0$ appears with the same sign as $-g_{00} > 0$ in the sum. Identifying $\Sigma_{\text{EM}} = -\ln(1 - f^2 A_0^2/|g_{00}|)$ completes the proof. □

Remark. *The electromagnetic entropy production Σ_{EM} is* not *independent of Σ_{grav}: it contains the factor*

$$\frac{f^2 A_0^2}{|g_{00}|} = f^2 A_0^2\, e^{\Sigma_{\text{grav}}}. \tag{47}$$

This means gravity amplifies the electromagnetic contribution to the total entropy production. Near a horizon ($\Sigma_{\text{grav}} \to \infty$, $|g_{00}| \to 0$), even a weak electromagnetic field produces large Σ_{EM}. This is the Σ-framework interpretation of the Wald–membrane paradigm: the horizon acts as an amplifier of information loss across all sectors.

Remark. *The decomposition $\Sigma_{\text{total}} = \Sigma_{\text{grav}} + \Sigma_{\text{EM}}$ is* additive in Σ, *which corresponds to* multiplicative *fidelities:*

$$F_{\text{total}} = e^{-\Sigma_{\text{total}}/2} = e^{-\Sigma_{\text{grav}}/2} \cdot e^{-\Sigma_{\text{EM}}/2} = F_{\text{grav}} \cdot F_{\text{EM}}. \tag{48}$$

Recovery from combined gravitational and electromagnetic information loss requires independent recovery of both sectors, with the total fidelity being the product of the individual fidelities.

B. α-running as Σ-structure

Remark (α-Running and Σ_{RG})**.** *The running of the fine-structure constant in QED is*

$$\alpha^{-1}(E) = \alpha^{-1}(E_0) - \frac{1}{3\pi}\,\Sigma_{\text{RG}}, \tag{49}$$

where

$$\Sigma_{\text{RG}} = 2\ln\left(\frac{E}{E_0}\right) \tag{50}$$

is the renormalization-group "entropy production" measuring the information lost by coarse-graining from UV scale E to IR scale E_0.

This has the same structural form as the gravitational entropy production $\Sigma_{\text{grav}} = 2\ln Q$ [2], with the identifications:

	Gravity	***RG Running***
Variable	$Q = 1/N_\phi$	E/E_0
Σ	$2\ln Q$	$2\ln(E/E_0)$
Information loss	*coarse-graining over geometry*	*coarse-graining over UV modes*
Fidelity bound	$F = e^{-\Sigma_{\text{grav}}/2}$	$\delta\alpha/\alpha \sim e^{-\Sigma_{\text{RG}}}$

The parallel is suggestive: in both cases, $\Sigma = 2\ln(\cdot)$ quantifies the "cost" of coarse-graining, whether the coarse-graining is over gravitational degrees of freedom (integrating out the environment to produce the effective lapse) or over UV quantum fluctuations (integrating out high-energy modes to produce the effective coupling).

This formal analogy does not, by itself, prove that α-running is gravitational in origin. However, it is consistent with the Σ-framework's central claim that all irreversibility—including the irreversibility of the RG flow—has a unified information-theoretic description.

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